

Evaluating Linguistic Proximity of Marathi as a Bridge Language for Konkani Translation

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Abstract—This paper investigates the linguistic and computational relationship between Marathi and Konkani, two sibling Indo-Aryan languages that share lexical, phonetic and orthographical similarities while differing in resource availability. Konkani remains a low-resource language which limits the effectiveness of neural machine translation (NMT) systems. To address this, we employ multiple linguistic similarity measures including orthographic edit distance, phonetic overlap and semantic embeddings to quantitatively establish Marathi's proximity to Konkani. Using Swadesh lists and parallel corpora, we compute similarity scores and evaluate zero-shot and finetuned translation performance with state-of-the-art models such as IndicTrans2. Experimental results demonstrate that Marathi achieves the highest alignment with Konkani across all metrics; with finetuned models utilizing Marathi yielded a BLEU-4 score of 32.95 and a chrF3 score of 72.02. This performance significantly surpasses other Devanagari script Indo-Aryan languages, with Hindi achieving a BLEU-4 of 26.15, followed by Maithili (24.59) and Nepali (23.70). Translation evaluations further confirm Marathi's role as an effective intermediate language for enhancing Konkani translation quality. These findings highlight the value of leveraging sibling languages in low-resource machine translation and provide empirical evidence for transfer learning strategies in the Indo-Aryan language family.

Index Terms—neural machine translation, low-resource languages, linguistic similarity, Indic languages.

I. INTRODUCTION

Developments in the field of neural machine translation (NMT) have significantly advanced the field of multilingual natural language processing, particularly for low-resource languages. In the context of Indo-Aryan languages, which often

share scripts and exhibit structural similarities, the performance of NMT models can vary based on the linguistic characteristics of the source and target languages [1], [2]. Konkani is an Indo-Aryan language spoken along India's

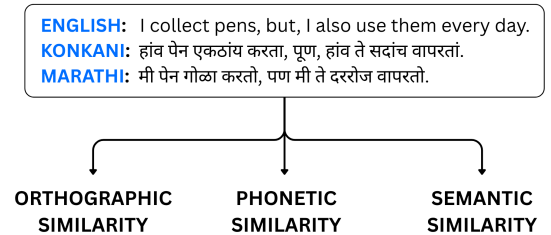


Fig. 1. Types of linguistic similarities measured between Marathi and Konkani

western coast; and despite its rich literary tradition and cultural significance, it remains a low-resource language underrepresented in computational linguistics [3]. It shares substantial linguistic similarities with Marathi, a widely spoken Indo-Aryan language which has more resources available for it. These similarities are found in their vocabulary, grammar, phonology, and usage of the Devanagari script, albeit certain distinguishing features maintain Konkani's status as a separate language from Marathi. The two are termed as 'sibling languages' and are related to each other as members of the Southern branch of the Indo-Aryan language family [15], sharing a common ancestor in Maharashtri Prakrit.

This close relationship suggests that transfer-learning from Marathi-pretrained models can be particularly effective for

English-to-Konkani translation. Studies in sentiment analysis and zero-shot learning confirm that models trained on Marathi corpora demonstrate high baseline performance on Konkani data due to their close linguistic relationship [4]. Recent advances in neural machine translation, notably the Helsinki OPUS-MT models pretrained on multiple Indic languages, have opened new possibilities for leveraging cross-lingual transfer to improve translation quality for low-resource targets [5], [6].

This paper aims to highlight the importance of using Marathi as an intermediate language towards Konkani translation by quantitatively establishing the linguistic similarity between Konkani and Marathi using multiple similarity metrics as displayed in Figure 1. We calculate and compare the scores of orthographic, semantic and phonetic similarity metrics for the two languages via Swadesh List. To further substantiate their close relationship, we analyze the Konkani translation performance of pretrained translation models. We use IndicTrans2 developed by AI4Bharat [7], which provides high-quality machine translation support from English to 22 scheduled Indic languages including Konkani. We utilize this model to perform both zero-shot translations and translations after being finetuned on the Konkani language. The target languages are chosen amongst those written using Devanagari script like Hindi, Marathi, Nepali and Maithili, and their similarity scores are evaluated with respect to Konkani for their respective machine translations. From this, we aim to empirically establish that Marathi's linguistic proximity to Konkani enables superior transfer and translation performance. While this work focuses on Konkani, our finding broadly applies to many low-resource languages that share close linguistic similarities with better-resourced languages, highlighting the value of leveraging such affinities to improve machine translation for underrepresented languages.

II. RELATED WORK

Multilingual NMT research more broadly has confirmed that adapting models from linguistically proximate high- or mid-resource languages consistently produces superior outcomes compared to relying on models trained on unrelated languages. In the machine translation domain, transfer learning strategies have been widely explored to leverage data from linguistically close source languages to improve translation for low-resource languages [8], [9]. Particularly for linguistically similar Indo-Aryan languages like Hindi and Marathi, studies prove that monolingual data and shared linguistic features can be exploited to address limited parallel data for similar language translation tasks [10], [11]. Here, the neural models use bilingual embeddings and linguistic features like POS tagging to improve the translation quality between closely related pairs, underscoring the benefits of transfer and multilingual learning within this family. A simple transfer learning approach showcases a 'parent' model trained on a high-resource language is fine-tuned on a low-resource pair by replacing only the training data, yielding a 'child' model that outperforms a baseline trained solely on low-resource data [12]. Employing

transfer learning for English-Marathi translation by appointing Hindi as a pivot language showed that linguistic proximity aids parameter sharing between models [13]. Moreover, initializing NMT models with pretrained parameters from linguistically related languages resulted in improved BLEU scores compared to unrelated parent models, highlighting higher accuracy in translation tasks in low-resource settings [14].

Linguistic similarity analyses between sibling languages have provided empirical foundations for such transfer strategies. Specifically studies measuring lexical, phonetic, orthographic, and semantic similarity scores between closely related Indo-Aryan pairs such as Marathi and Konkani have reported high overlap and shared linguistic characteristics. This confirms their close genetic and typological relationship [17]–[19]. Carrying out sentiment analysis of a model fine-tuned on Marathi gives high accuracy when tested on Konkani sentences despite not being explicitly trained on Konkani data [16]. This outcome highlights how effective zero-shot learning is, where the model generalizes well across languages that are linguistically similar. Such quantitative evidence supports the hypothesis that sibling languages with substantial lexical and phonetic proximity enable more effective cross-lingual model adaptation and resource sharing and yield improved translation quality in low-resource settings.

III. PROPOSED METHODOLOGY

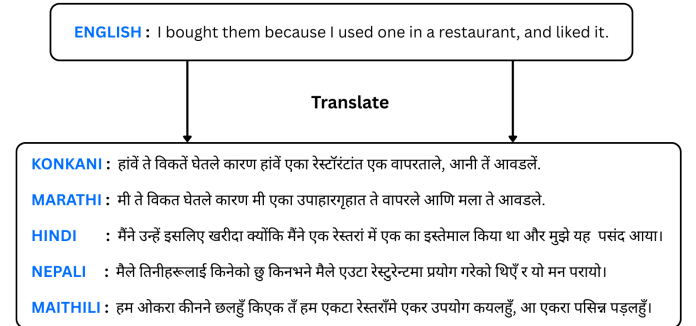


Fig. 2. Translation of a common English sentence into various Indo-Aryan languages across different branches that use Devanagari script

Using the IndicTrans2 model, we aim to establish Marathi as a useful intermediate language towards Konkani translation by establishing the linguistic closeness between these two languages. This is illustrated by using measuring the similarity in their orthographical and phonetic and nature. Orthographical similarity is a measure of the similarity between the lexicon and morphology of these languages by means of orthography, ie, the manner of representation of words. Phonetic similarity measures the similarity in the actual pronunciation of these words, which coupled with orthographical similarity gives a more complete insight into comparing lexica of two languages. These 2 metrics are measured by considering a Swadesh list.

We further highlight the usefulness of Marathi as an effective language to perform transfer-learning for Konkani by comparing zero-shot and finetuned machine translations of sentences into multiple Indo-Aryan languages against a Konkani

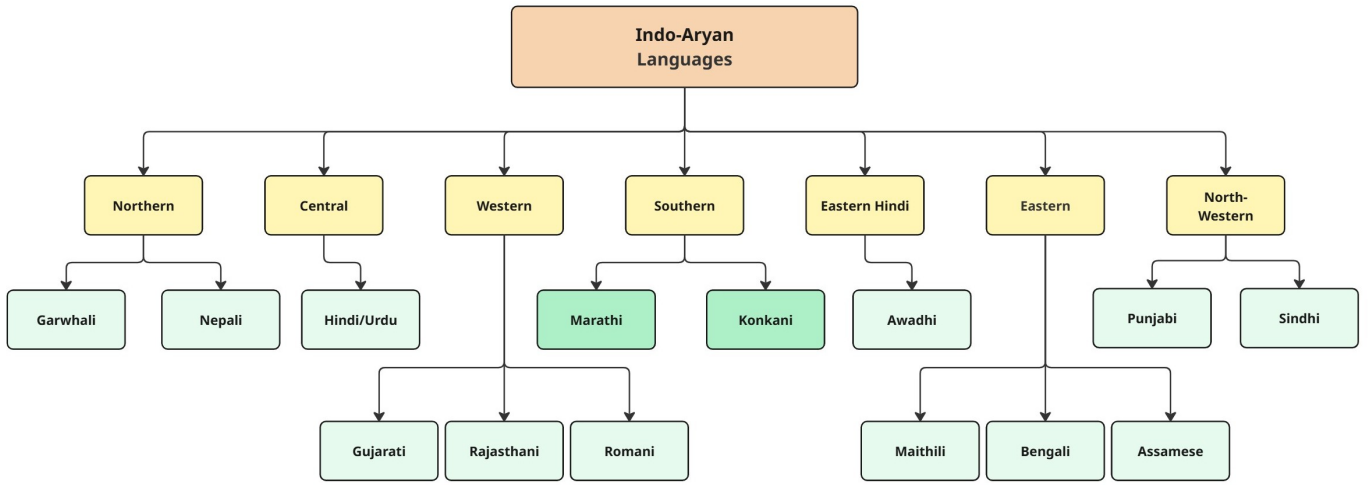


Fig. 3. Classification of Marathi and Konkani within the Indo-Aryan language family

reference. Figure 3 represents the hierarchy of relation between Indo-Aryan languages. We have chosen languages specifically using the Devanagari script taken from different branches of the Indo-Aryan language family: Marathi for the Southern group, Hindi for the Central Group, Nepali for the Northern Group and Maithili for the Eastern Group. Although languages like Kannada may be similar to Konkani due to geographical proximity and loan words, it does not share the same script as Konkani. Figure 2 shows a sample translation from English to the respective languages. Here, semantic similarity is chosen as an additional metric to assess the similarity in the intended meaning of the translations, acting as a measure for the preservation of meaning between Konkani and each of these Indo-Aryan languages.

A. Swadesh List

Invented by the comparative and historical linguist Morris Swadesh [30] as a means to compare languages and discover relationships between them, Swadesh lists are a compilation of concepts considered universal across human languages such that any language would have a term for it. This is presented in the form of a list of translations for these concepts in each considered language, placed side-by-side for comparison. Such manner of presentation makes these lists useful for comparative linguistics and glottochronology, which aim to study the presence and nature of relationships between different languages.

Due their universality across languages, we use a comparative Swadesh list containing English, Konkani and Marathi to analyze the orthographical and phonetic similarities between the two Indo-Aryan languages in consideration, with English used as an intermediate non-Indo-Aryan language which provides a neutral base to understand the listed terms.

B. Model Architecture

IndicTrans2 is a group of open-source machine translation models designed for rapid multilingual deployment [7]. The

original IndicTrans2 model supports all 22 scheduled Indian languages, and the IndicTrans2-M2M model supports direct translations between these 22 languages. This study utilizes the specific IndicTrans2-en-indic-1B model.

The IndicTrans2 model uses a transformer-based architecture [20] with 18 layers for both encoder and decoder and 16 attention heads with all modules pre-normalized [21]. GELU activation is also used [22]. The model contains a total of 1.1B parameters.

C. Model Pretraining

The IndicTrans2 model has been trained on the the Bharat Parallel Corpus Collection (BPCC) dataset which contains a total of 230M bitext pairs. These pairs are sourced existing collections such as Samanantar [23] and NLLB [24] as well as new pairs mined from various sources such as NPTEL and *archive.org*. Additionally, seed training data has also been used as part of this dataset containing human translations of English sentences to various Indic languages.

This data is then preprocessed with standard methods, strict data deduplication and script unification. Some additional preprocessing is applied to handle special cases such as emails, dates and URLs. Subword-level Tokenization [25] is then applied with two separate tokenizers that use SentencePiece library [26] and byte-pair-encoding (BPE) algorithm. Finally, source and target language tags are prepended to indicate the translation direction [27] [28] [29]. With FLORES-200 [24] multi-domain development set used for validation, IndicTrans2 is then trained in two phases: auxiliary training and downstream training.

IV. EXPERIMENTATION

A. Experimental Setup

1) *Datasets*: We utilize the publicly available sea-rod/konkani-marathi-sentiment-dataset consisting of parallel English, Marathi and Konkani sentences for sentiment analysis [16]. The semantic similarity using

IndicBERT between the Marathi and Konkani paired translations came out to be 0.9061. To ground lexical similarity analysis in core vocabulary, we use Swadesh lists for both languages obtained from Wikipedia: the Marathi Swadesh list and the Konkani Swadesh list. These lists provide a set of stable, basic vocabulary essential for comparative linguistic studies.

2) *Models:* We make use of ai4bharat/indictrans2-en-indic-1B zero-shot for translation. The language codes used as per the IndicTrans usage are as follows:

- English - eng_Latn
- Konkani - gom_Deva
- Marathi - mar_Deva
- Nepali - npi_Deva
- Maithili - mai_Deva

We performed experimentation on a Tesla L4 GPU with the max_length was set to 128. The batch size was 64 for zero-shot translation. For the finetuning process we chose a batch size of 16 and trained the model for 15 epochs with a learning rate of 3e-5, and weight decay of 0.01.

TABLE I
SIMILARITY SCORES FOR MARATHI AND KONKANI ON THE SWADESH
LISTS

Similarity Metric	Average Score
Bigram Similarity	0.2233
Trigram Similarity	0.2507
Phonetic Similarity	0.8690
Orthographic Levenshtein	0.3747
Orthographic SeqMatch	0.4329

3) *Evaluation Metrics:* To show the similarity between Marathi and Konkani, we evaluate orthographic and phonetic similarity for the Swadesh lists.

- Orthographic similarity is computed based on normalized edit distance using the Levenshtein distance and sequence alignment measures applied to Devanagari-normalized word forms.

The Levenshtein distance between two strings a and b of lengths $|a|$ and $|b|$, denoted as $\text{lev}(a, b)$, is defined recursively as:

$$\text{lev}(i, j) = \begin{cases} i & \text{if } j = 0 \\ j & \text{if } i = 0 \\ \text{lev}(i-1, j-1) & \text{if } a_i = b_j \\ 1 + \min \begin{cases} \text{lev}(i-1, j) \\ \text{lev}(i, j-1) \\ \text{lev}(i-1, j-1) \end{cases} & \text{otherwise} \end{cases} \quad (1)$$

We normalize it to obtain the orthographic similarity:

$$\text{OSim}_{\text{Lev}}(a, b) = 1 - \frac{\text{lev}(a, b)}{\max(|a|, |b|)} \quad (2)$$

The sequence matching similarity $\text{OrthSim}_{\text{Seq}}$ is computed as the ratio of matched subsequences between normalized strings:

$$\text{Psim}_{\text{Seq}}(a, b) = \frac{2 \times M}{|a| + |b|} \quad (3)$$

where M is the length of the longest matching subsequence.

- Phonetic similarity is approximated using counts of vowels v_i , consonants c_i , and lengths l_i of the Devanagari-normalized words:

$$\text{Psim}(a, b) = \frac{1}{3} \left(1 - \frac{|v_1 - v_2|}{L} + 1 - \frac{|c_1 - c_2|}{L} + 1 - \frac{|l_1 - l_2|}{L} \right) \quad (4)$$

where $L = \max(l_1, l_2, 1)$ to avoid division by zero.

- Additionally, we report n-gram similarities for bigrams and trigrams defined as the Jaccard similarity over the sets of character n-grams:

$$\text{JSim}_n(a, b) = \frac{|G_n(a) \cap G_n(b)|}{|G_n(a) \cup G_n(b)|} \quad (5)$$

where $G_n(s)$ is the set of character n -grams extracted from string s .

To complement the above similarity scores, we also evaluate machine translation quality and semantic closeness between Marathi and Konkani using standard metrics :

- **BLEU-4** : This measures n-gram overlap precision between the candidate and reference translations.
- **chrF** : Character n-gram F-score capturing subword similarity.
- **TER** : Translation Edit Rate capturing necessary edits between hypothesis and reference.
- **BERTScore** : Embedding based similarity using contextual embeddings to assess semantic fidelity. It is calculated using ai4bharat/indic-bert.
- **Semantic Similarity** : Embedding based similarity using sentence-transformers/LaBSE (language agnostic BERT sentence embeddings).

These metrics collectively provide a comprehensive evaluation of linguistic similarity between Marathi and Konkani from surface orthographic resemblance to deep semantic equivalence.

V. RESULTS

A. Linguistic Similarity Analysis on Swadesh Lists

Table I presents the average similarity scores for Marathi and Konkani computed on the Swadesh word lists using multiple orthographic and phonetic metrics. Notably, the highest score is observed for phonetic similarity (0.8690), showing a significant overlap in the sound structures of the basic vocabularies of both languages. Orthographic similarity metrics calculated via Levenshtein distance and sequence matching, yield moderate scores of 0.3747 and 0.4329 respectively,

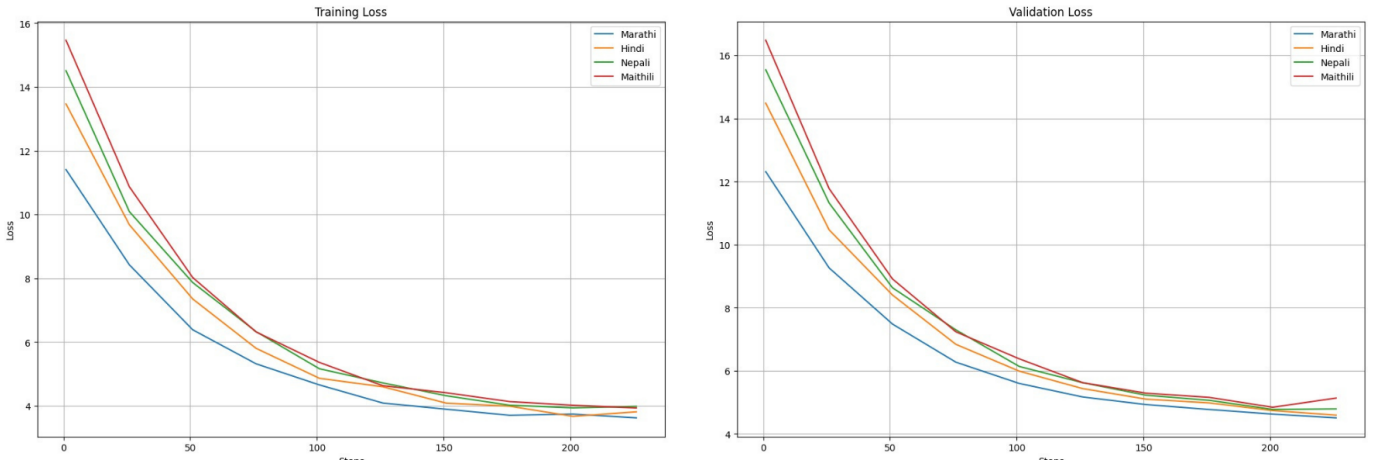


Fig. 4. Training and Validation losses of Marathi, Hindi, Nepali and Maithili across 225 steps.

which reflects surface-level spelling resemblance but also the presence of systematic orthographic variation. In contrast, bigram and trigram similarity metrics register lower values (0.2233 and 0.2507), suggesting that while many words are similar in overall sound and form, short subword overlapping patterns are less frequent.

B. Translation Metrics with Konkani References

Table II summarizes translation performance for Marathi, Hindi, Nepali, and Maithili sentences evaluated against Konkani references using BLEU-4, TER, chrF3, BERTScore, and semantic similarity metrics. Marathi achieves the highest pretrained translation quality with a BLEU-4 score of 1.2671, chrF3 of 30.302, BERTScore of 0.694 and semantic similarity of 0.656, surpassing all other Devanagari languages tested. The TER (Translation Edit Rate) for Marathi (93.242) is substantially lower than for Hindi, Nepali, and Maithili, further emphasizing translation proximity. These results are both improved and further verified by the finetuned translation, as evidenced by its score: Marathi achieves BLEU-4 score of 32.949, chrF3 of 72.016, BERTScore of 0.888 and a further reduced TER value of 43.212, maintaining itself as the lowest among the Indic languages considered. Additionally, Figure 4 displays the training and validation losses for each language. Loss for Marathi starts at a lower value and converges faster, indicating similarity between existing Marathi and learned Konkani representations.

In contrast, Hindi, Nepali, and Maithili display lower translation scores, with BLEU-4 values below 0.5 for pretrained and below 30 for finetuned scores, and chrF3 and semantic similarity metrics consistently trailing behind Marathi. BERTScore and semantic similarity consistently corroborate the pattern of greater linguistic closeness between Marathi and Konkani compared to the other Indo-Aryan languages examined.

VI. CONCLUSION AND FUTURE WORK

This paper investigated the linguistic proximity between Marathi and Konkani by combining orthographic, phonetic,

TABLE II
TRANSLATION METRICS FOR MARATHI, KONKANI, HINDI, NEPALI AND MAITHILI WITH RESPECT TO KONKANI REFERENCE SENTENCES

Language	BLEU-4	TER	chrF3	BERTScore	Semantic
Pretrained					
Marathi	1.2671	93.242	30.302	0.694	0.656
Hindi	0.424	135.685	18.281	0.654	0.621
Nepali	0.178	108.498	16.543	0.687	0.636
Maithili	0.333	124.368	16.933	0.689	0.639
Finetuned					
Marathi	32.949	43.212	72.016	0.888	0.868
Hindi	26.154	45.326	68.987	0.879	0.861
Nepali	23.700	44.404	68.477	0.870	0.844
Maithili	24.594	70.055	68.707	0.876	0.855

and semantic similarity metrics applied to Swadesh lists and parallel corpora. The analysis quantitatively confirmed that Marathi shares a high degree of linguistic similarity with Konkani especially in phonetic structure. Experimental translation evaluations using the IndicTrans2 model further demonstrated that Marathi serves as an effective intermediate language for enhancing Konkani translation quality in comparison to Devanagari-script Indo-Aryan languages like Hindi, Nepali, and Maithili. These findings highlight the value of leveraging linguistic relationships of sibling language and transfer learning strategies to improve machine translation for low-resource languages like Konkani.

For future work, we aim to explore more advanced cross lingual transfer methods. This will include fine-tuning multilingual models specifically on Marathi-Konkani aligned corpora. Our goal is to improve translation performance. Extending this approach to other low-resource language pairs within the Indo-Aryan family could validate the generality of our findings. Additionally, investigating unsupervised or semi-supervised approaches may help overcome resource scarcity and facilitate translation model development for many underrepresented Indic languages.

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